Wafer thin laptops with 8 hours of battery life

It isn't science fiction, it's science fact!

Did you always wish your laptop to be a tad lighter, a bit faster, its battery to last a few hours more? The tech gods are surely smiling down on you, and the rest of us demanding lot, for there's a new kid on the block that will fight tablets for all the tech headlines in 2012. Ultrabooks, essentially thin-and-light laptops with improved performance chops, will go mainstream this year.

The world beyond netbooks

When the mini-laptop PC captured popular imagination, back in 2007, Intel had only coined the word 'netbook' which got attached to the early mini-notebooks launched subsequently thereafter. Aimed at the entry-level of a price-sensitive notebook market, netbooks were never meant to be processing powerhouses; they never really gave traditional 13-inch, 14-inch laptops a run for their money. Netbooks were a huge hit for the past 3-4 years mainly because of their inexpensiveness and thin-n-light form factor. These two ingredients are central among the new breed of thin, powerful ultrabook laptops that are going to launch left, right and centre this coming year.

The Ultrabook cometh

The industry is betting big on the ultrabooks to take off. According to an Intel prediction, by the end of 2012, 40 per cent of the notebook market will comprise of ultrabooks. That's just astonishing, if true, as ultrabooks only started to appear late last year (we reviewed the Acer Aspire S3 back in November). For Intel to project such a figure means they have all the prominent OEMs on board for this to happen. And not surprisingly, some of the major players in the notebook market have either launched or are in the process of launching their respective ultrabook line. We know ASUS, Acer, Lenovo and Samsung have already launched their ultrabook offerings in the Indian market. We expect brands like Dell, HP, Sony, MSI, Fujitsu, Toshiba and others to launch their branded ultrabooks very soon on Indian shores, too.

PMP IS DEAD

With the advent of smartphones, which couple up as efficient media players, the PMP segment is almost at the end of the road. Let's be honest, the best PMP in our books, the Apple iPod Touch, was released over a year ago. Worthy contenders are fast disappearing. Our advise: hold on to your beloved PMP from the pre-smartphone era, for come the end of this year, they will (in all probability) be relegated to history.

More processing chops, less power consumption

While most of the ultrabooks launched early this year will continue to sport existing Sandy Bridge chips, new hardware platform will take processing and computing to the next level. Intel's 22nm sophistication of Sandy Bridge, codenamed Ivy Bridge, will finally start shipping towards the end of this year. The most path-breaking difference it will provide compared to existing Sandy Bridge processors is its innovative Tri-gate transistor design — it gives the same performance as the current chips having 2D transistors at a 50 per cent lower power. The upcoming Ivy Bridge chips overall push existing Intel CPU performance up by 20 per cent and onboard graphics output is higher by a whopping 30 per cent. All things said and done, despite any major breakthroughs in Li-ion batteries, we expect the average thin-n-light laptop (or ultrabooks) launched this year to last well over 5 hours, some may even touch the coveted 8-hour mark hailed by early netbooks. Imagine having a thin 13-inch ultraportable laptop that doesn't burn a hole through your pocket, performs as well as any mainstream 15-inch laptop and lasts for the better part of your work day. Now wouldn't that be something to cheer about?

> "Due to the rapidly growing amounts of data being produced, management and easy accessibility of data from anywhere will become crucial for private and professional users."
>
> Khalid Wani, Sales Director, Western Digital

Intel, AMD roadmap

Not just this, the next generation of Intel Atom chips dubbed Cedar Trail should start shipping early this year, boosting netbook performance. Equipped with Cedar Trail Atom chips, N2800 and N2600, netbooks launching this year will better handle 1080p video, HDMI and DisplayPort out, and also an onboard GPU with DirectX 10.1 support. AMD has its own few thin-n-light platform updates planned around the same time. The next level of Brazos chips will arrive towards the end of this year that will further open up the thin-n-light laptop segment. Krishna is scheduled to replace the existing AMD Fusion "Zacate" APU, while Wichita is the next level of "Ontario" APUs for use in laptops and netbook-like devices. AMD APU-based thin-n-light laptops or ultrabook competitors may sport AMD Radeon HD 7000 graphics. d